1 Density of states

1.1 Python-simulated DOS

The real-space DOS shows a significant peak at $E = 2t$, representing the most generate energy state.

1.2 Comparison with Experimental Data

We compared our results with the experimental DOS (dI/dV) of KV_3Sb_5[1]. The range of energy for KV_3Sb_5 in the d_z^2 orbital is around 0.5eV, comparing with our simulation, which has the range 6t, therefore, t=0.083eV. The equivalent peak using our simulation for the experimental data is therefore 2t=1.7eV. The experimental date shows a peak E=-0.1eV. The difference between our simulation prediction and the actual result could be explained through difference in orbitals. However, the general trajectory of one dominant peak agrees with our simulation.

1.3 Band structure

Our python-simulated band structure is the same as the theoretical result, up to a reflection. This reflection is explained by the fact that we used GKM path, yet the paper used GMK path. However, do to limitation of ideal models, the experimental result is more complicated the our simulation. However, we could still see the characteristic of a plat band with curvature.

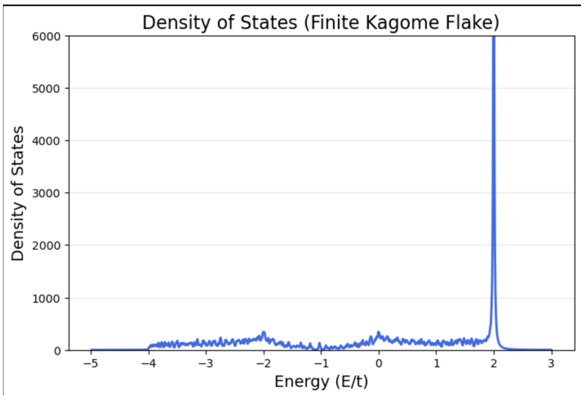

Figure 1: Python simulated Local Density of States for $t = 1$.

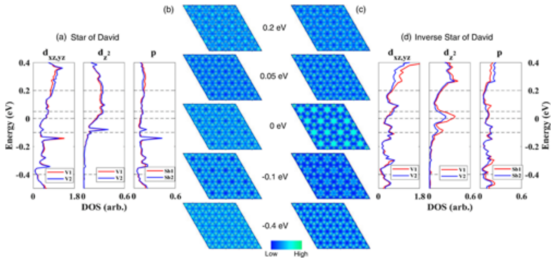

Figure 2: Experimental DOS for KV_3Sb_5.

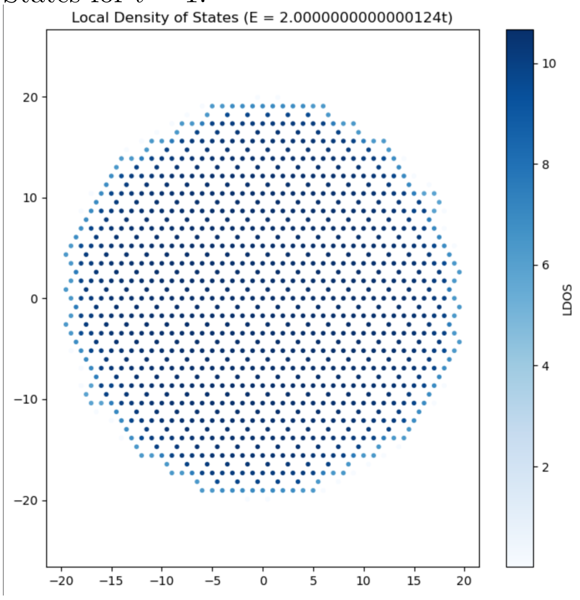

Figure 3: Python simulated Local Density of States at E=2t

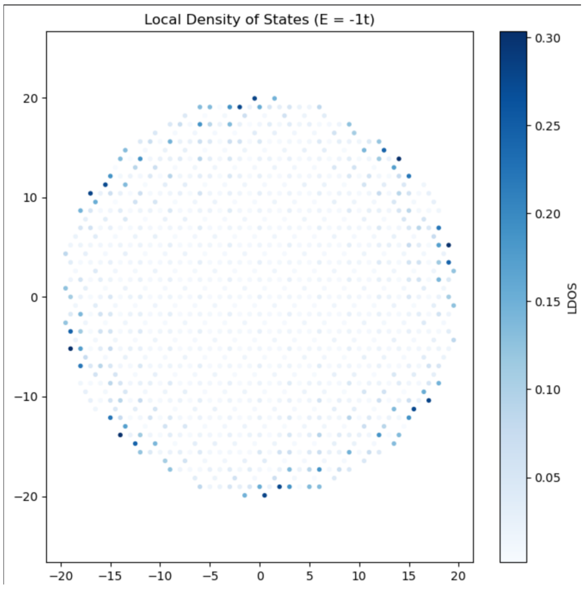

Figure 4: Python simulated Local Density of States for $E = -t$.

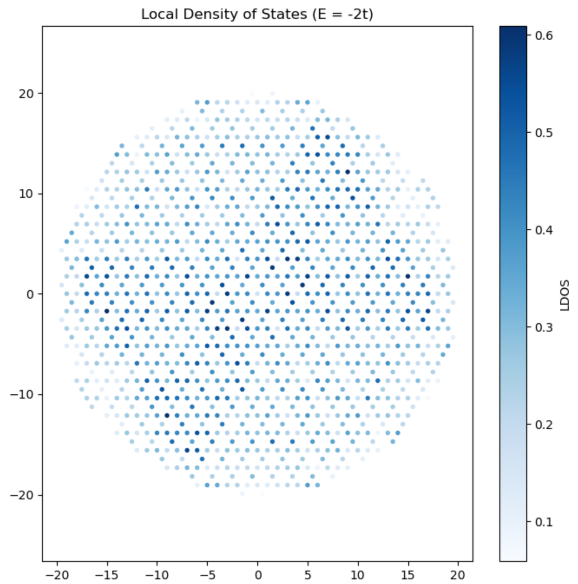

Figure 5: Python simulated Local Density of States at E=-2t

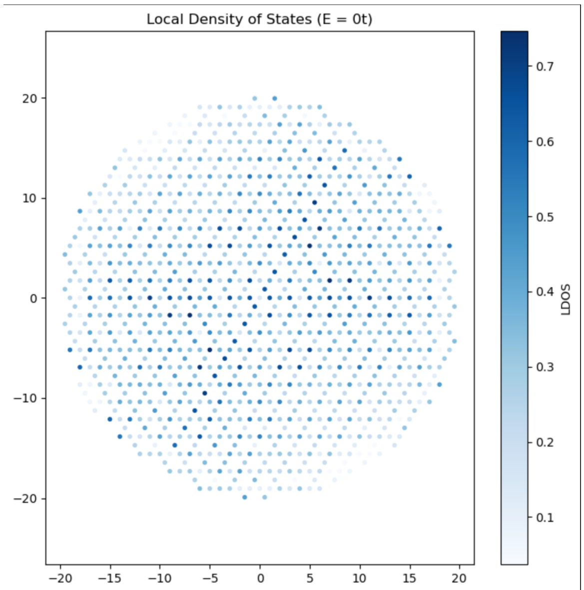

Figure 6: Python simulated Local Density of States at E=0

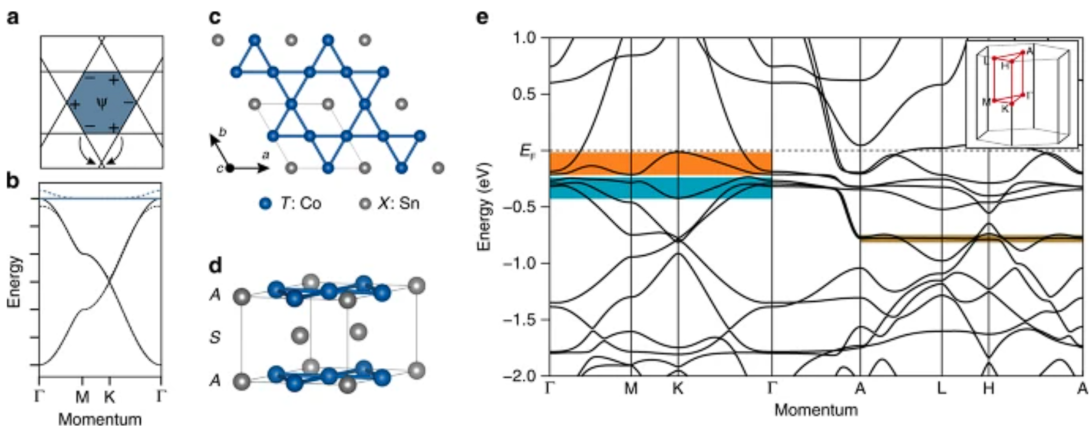

Figure 7: Theoretical origin of the flat band and its realization in CoSn. (a) Real-space destructive interference confining an electron in a kagome hexagon. (b) Ideal tight-binding band structure showing the corresponding flat band. (e) DFT band structure of CoSn highlighting the flat band domains. Figure adapted from Kang *et al.* [2].

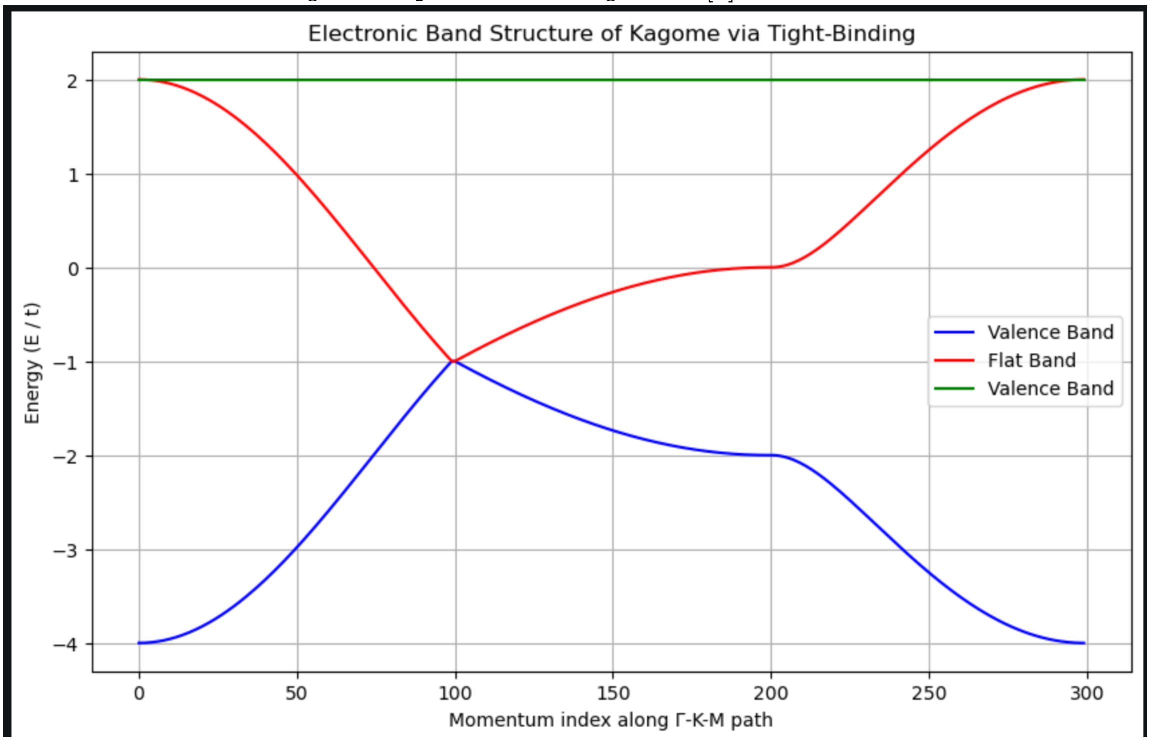

Figure 8: Python simulated energy band structure

1.4 Next-Nearest-Neighbor at t'=0.1t

From the below simulation, a clear and dominant pattern is that the NNN interactions is reducing the degeneracy of E=2t. From the LDOS simulation, we can conclude that the main source of such reduction is the atoms at the boundary of our kagome flake, where the density of states for E=2t decreased the most. This has a heavy influence on our global density of states for E=2t, as shown in figure 6. Finally, such reduction in degeneracy caused minor fluctuations to our flat band, which is more similar to the experimental result in figure 4e [2]. Therefore, we conclude that including the next-nearest-neighbor interactions with t'=0.1t is a better model compared to nearest neighbor tight binding.

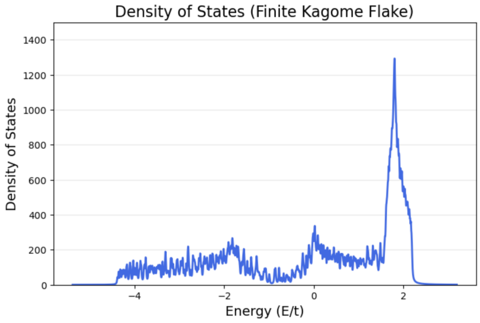

Figure 9: Local density of state after introducing NNN interactions

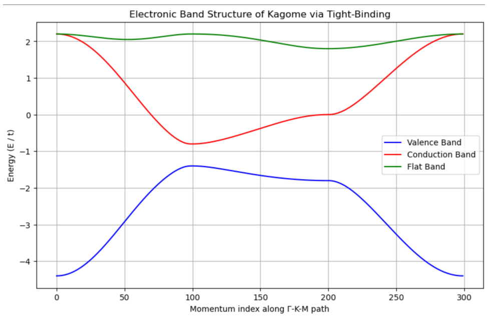

Figure 11: Energy band structure after including NNN interactions

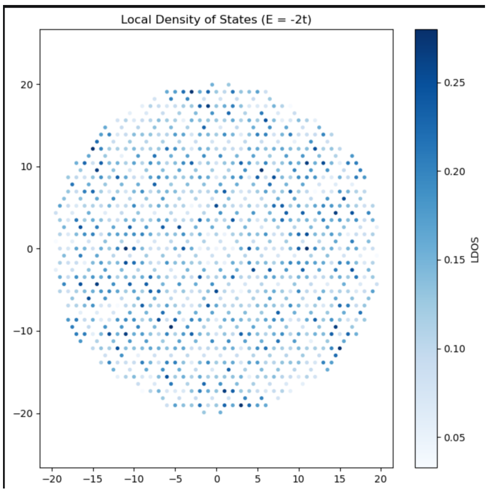

Figure 13: Local density of state at E=-2t after introducing NNN interactions

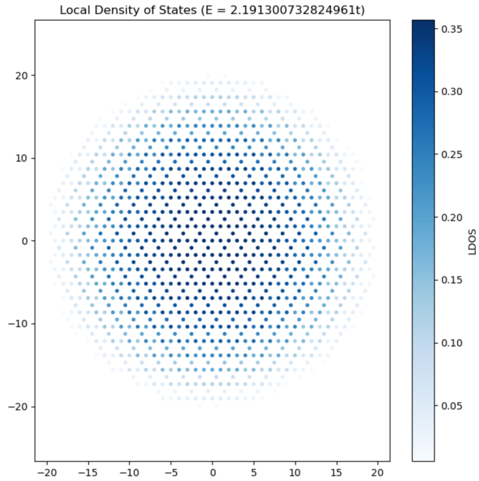

Figure 10: Global density of states after including NNN interactions

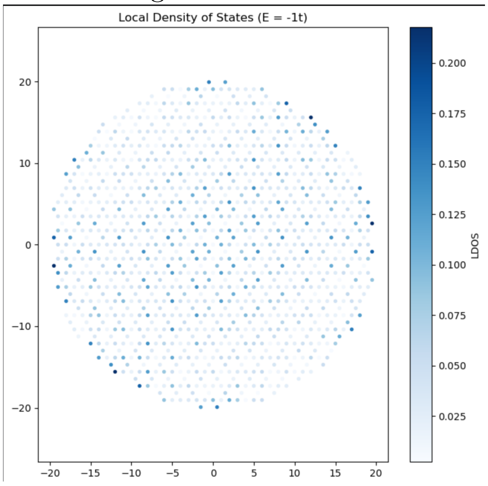

Figure 12: Local density of state at E=-1t after introducing NNN interactions

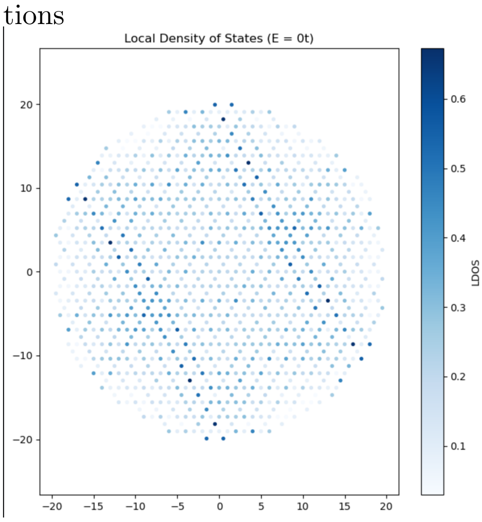

Figure 14: Local density of state at E=0 after introducing NNN interactions

References

[1] Hengxin Tan, Yizhou Liu, Ziqiang Wang, and Binghai Yan. Charge density waves and electronic properties of superconducting kagome metals. *Phys. Rev. Lett.*, 127:046401, Jul 2021.

[2] Mingu Kang, Shiang Fang, Linda Ye, Hoi Chun Po, Jonathan Denlinger, Chris Jozwiak, Aaron Bostwick, Eli Rotenberg, Efthimios Kaxiras, Joseph G. Checkelsky, and Riccardo Comin. Topological flat bands in frustrated kagome lattice cosn. *Nature Communications*, 11(1):4004, Aug 2020.